

Nadhir Halbouni

Software Engineer

✉ nadhir.halbouni5@gmail.com

☎ +216 50 058 735

🏠 Ariana - Tunisia

🐙 [n3dhir](https://github.com/n3dhir)

🌐 nadhir-halbouni/

🌐 n3dhir.com

Summary

Full-stack engineer specializing in **data-heavy SaaS systems**, **backend reliability**, and **automation**. Experienced in building and scaling production systems across multiple repositories, handling real-time features, payments, and growth mechanisms. Strong focus on **ownership**, **system design**, and **delivering end-to-end solutions** from database to user experience.

Experience

Drivago

Ariana, Tunisia

Software Engineer

September 2024 - Present

- Designed and built a **cross-repository training platform** spanning 7+ services, enabling secure multi-device usage and supporting concurrent usage without session conflicts.
- Integrated **payment gateways (Konnect, Flouci)**, implementing full payment lifecycle and automating subscription billing across thousands of transactions.
- Owned **production incident resolution** for 500+ client agencies, reducing incident resolution time by rapidly reproducing and debugging cross-platform issues.
- Managed **production deployment** on **Linux servers** using **Nginx and PM2**, maintaining uptime and overseeing environment configuration across all services.
- Engineered **automation systems (cron jobs)** for scraping, database backups, and recurring billing, eliminating manual operations and saving hours of weekly maintenance work.
- Developed **authentication and security systems**, including magic-link login, OAuth (Google/Facebook), and real-time fraud detection enforcing single-session access.
- Led migration of 4 core repositories from Angular v14 → v19, standardizing architecture and enabling full RTL/LTR support.
- Built a **referral & coupon system** driving user acquisition and retention through modular discount logic and reward tracking.

Drivago

Ariana, Tunisia

Software Engineer Intern

July 2024 - August 2024

- Built **BI dashboards** tracking SaaS metrics (MRR, ARR, churn, ARPU) to support product decisions.
- Designed and optimized **PostgreSQL queries** for retention analysis and cohort tracking.
- Developed advanced **segmentation & filtering systems** for backoffice operations (subscriptions, payments, activity).

Projects

Testify — Technical Assessment Platform

November 2023 - June 2024

Web-based technical assessment platform featuring code execution, automated evaluation, and backend processing for candidate submissions.

- Built a full-stack **technical assessment platform** supporting MCQ, coding challenges, and open-ended evaluations.
- Integrated a **C++ code execution api** to run user-submitted code, with test case validation, automated scoring, and pass/fail evaluation.
- Designed backend evaluation pipeline handling **submissions, validation, and result aggregation**.
- Tech: Angular, Node.js, Express.js, MongoDB, ACE Editor

Live Demo: testify.n3dhir.com

Demo Video: drive.google.com/file/d/11PjLQudEmKk0j9YbJ3aseVqYhDcJsMyp

Skills

- **Core Strengths** Backend Systems, API Design, REST APIs, Authentication & Security, Payments Integration, Automation, Data Modeling
- **Backend & Data** Node.js, Express.js, PostgreSQL, Knex.js, MongoDB
- **Frontend** Angular (v14–v19), RxJS, Angular Material, SCSS
- **Infrastructure & Tools** Linux, Nginx, PM2, Cron Jobs, ngrok, Git/GitHub
- **Other** SSR, SSG, SEO Optimization, OAuth (Google/Facebook)

Education

Software Engineering Degree 2023–2026
Higher Institute of Informatics (ISI), Ariana, Tunisia

Computer Science Degree 2020–2023
Higher Institute of Informatics (ISI), Ariana, Tunisia

Certifications

- | | |
|---|----------------|
| • <u>SQL (Advanced) Certificate — HackerRank</u> | June 2023 |
| • <u>Problem Solving (Intermediate) Certificate — HackerRank</u> | August 2022 |
| • <u>Relational Database — freeCodeCamp</u> | June 2023 |
| • <u>JavaScript Algorithms and Data Structures — freeCodeCamp</u> | June 2023 |
| • <u>Algorithmic Toolbox — Coursera</u> | May 2022 |
| • <u>Meta Hacker Cup 2022</u> | September 2022 |

Languages

Arabic (Native), English (Professional), French (Professional), German (Basic)